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Abstract This paper deals with transient stability analysis of a multi-machine power system using direct methods based on the transient energy of the system when subjected to a large disturbance. The component of the transient energy that does not contribute to system separation is identified, and is corrected for in the computation of the critical transient energy at clearing. The technique is validated by computing critical clearing times for two test systems and comparing the results with those obtained by time solutions.

INTRODUCTION

This is the second of two papers dealing with the use of direct methods in transient stability analysis of multi-machine power systems. In the first paper [1], the trajectory of the system subjected to a large disturbance is analyzed in considerable detail. Valuable information is obtained concerning the concept of controlling unstable equilibrium point (u.e.p.) for the particular transient analyzed, the manner in which some machines tend to lose synchronism, and the components of the system's transient energy. The components of the transient energy that are directly responsible for system separation are thus clearly identified.

In this paper, the information and conclusions obtained in the companion paper are used to: 1) identify the correct critical transient energy for the system, and 2) validate the results by computing critical clearing times for two test systems.

The Transient Energy

The equations for the system's transient energy are discussed in [1]. The transient energy function used is that proposed in [2] and is repeated here for convenience.

At a given instant where the generator angles with respect to the inertial center are θ_i , $i=1,2,\dots,n$, the system energy function is given by

$$V = \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \sum_{i=1}^n P_i (\theta_i - \theta_i^s) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left[C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s) + I_{ij} \right] \quad (1)$$

where

- θ^s = stable equilibrium angle
- M_i = inertia constant of machine i
- ω_i = speed of machine i with respect to inertial center

P_i = mechanical power-equivalent shunt load for machine i

C_{ij}, D_{ij} = constants that include machine internal voltages and network admittances

I_{ij} = an approximated term to account for the transfer conductances

$$= D_{ij} \frac{\theta_i + \theta_j - \theta_i^s - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s) \quad (2)$$

Since the transient energy is computed at a given instant (defined by θ, ω) with respect to a reference point, the following notation will be used

$$V \Big|_{\theta^a}^{\theta^b} = \text{transient energy at } b \text{ with respect to } a \quad (3)$$

with the appropriate parameters used in Eqs. (1) and (2).

The Test Networks

The same test networks used and described in [1] are used here: 1) the 4-generator test system, and 2) the 17-generator reduced Iowa network. These two power networks are shown in Figs. 1 and 2.

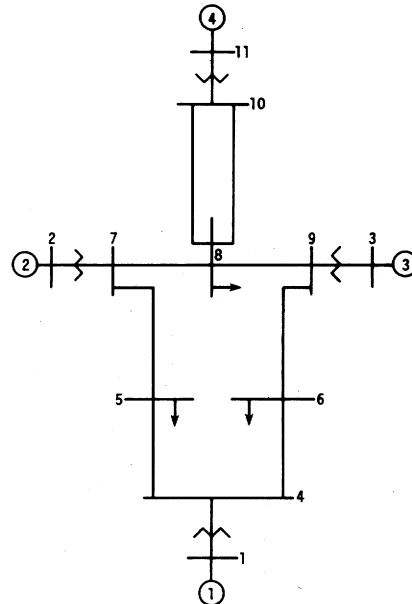


Fig. 1. The 4-generator test system.

CRITICAL TRANSIENT ENERGY

The assumptions commonly made in the literature on direct methods of transient stability analysis are [3]:

- The critical transient energy that the system can have at the beginning of the post disturbance period is equal to the potential energy at the u.e.p., V_u .
- At fault clearing, the system transient energy cannot exceed that

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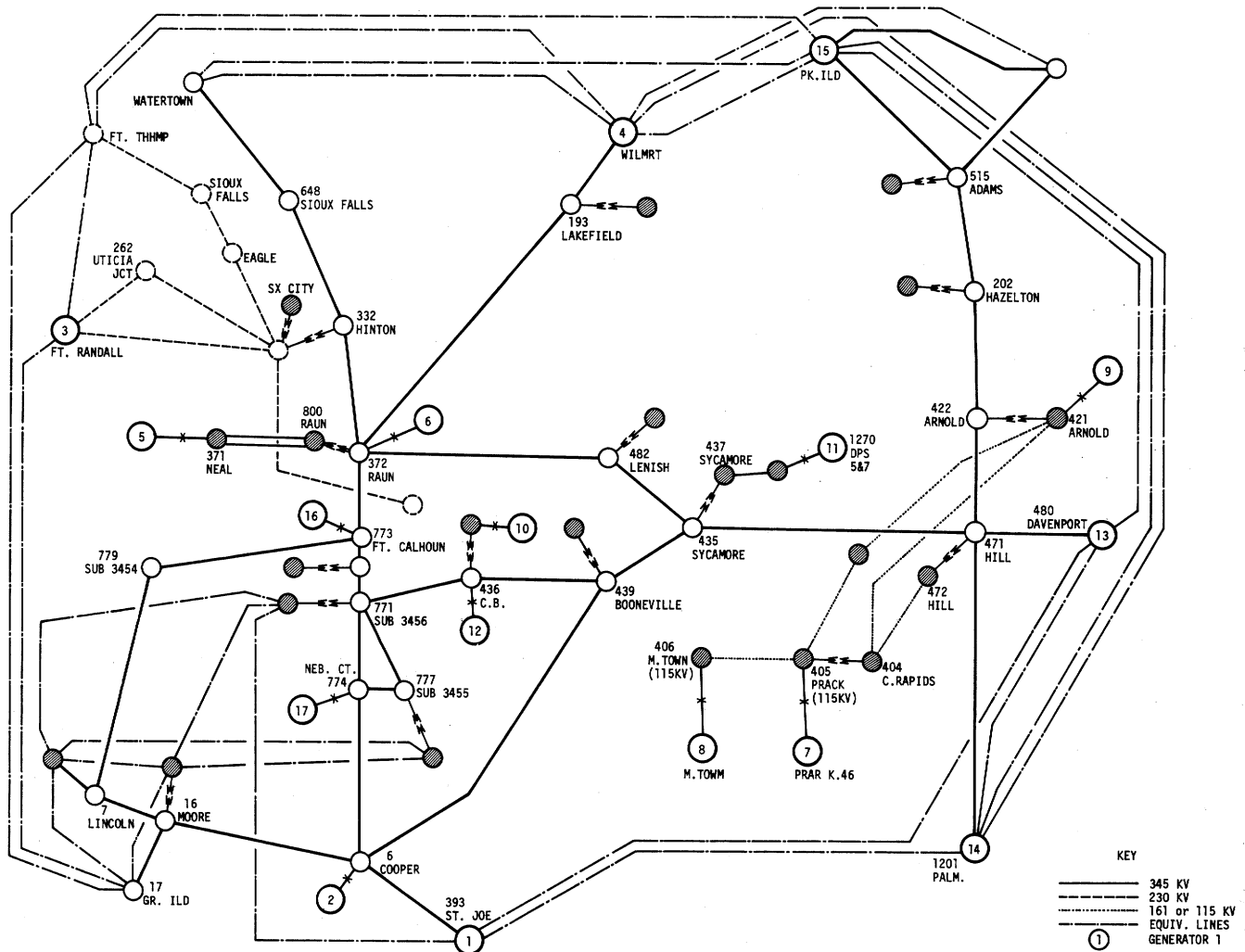


Fig. 2. 17-generator system. (Reduced Iowa System)

of V_u .

The companion paper [1] raised the following issues:

1. The ability of the post disturbance network to absorb (or convert) the transient energy at fault clearing θ^c (overlooking the issues raised below momentarily) will depend upon the expression $V_{cl}^{\theta^c}$. If the energy at θ^c and at θ^u are obtained separately, as is usually the case in transient stability analysis, they must be computed to the same reference point. However, usually $V_{cl} = V_{cl}^{\theta^{s1}}$, while $V_u = V_{cl}^{\theta^u}$. Thus, a correction must be introduced if θ^{s1} , the prefault stable equilibrium point, differs from θ^{s2} , the post fault stable equilibrium point.
2. When the potential energy (P.E.) is maximum and the critical generators reach their maximum swing, the system kinetic energy (K.E.) is usually *not* zero. Thus, not all of the kinetic energy contributes to system separation. The kinetic energy that does not contribute to instability need not be converted to potential energy for stability to be maintained. This component must be identified and corrected for in computing the critical transient energy.

Correction Due to Change in θ^s

The manner in which transient energy is usually computed at clearing is given by

$$V_{cl} = V \Big|_{\theta^{s1}}^{\theta^c}$$

while at θ^u

$$V_{cr} = \Big|_{\theta^{s2}}^{\theta^u}$$

Using the same reference for both is equivalent to adding a correction term to the energy at θ^u

$$V_{\text{correction}} = V \Big|_{\theta^{s1}}^{\theta^{s2}} \quad (4)$$

4-generator system

Pertinent data is shown in Table I for a three-phase fault at bus 10 cleared by opening one of lines 8-10.

TABLE I
4-Generator System
Fault at Bus 10, Line 8-10 Cleared

Generator Number	θ^{s1} deg	θ^{s2} deg	θ^u deg
1	6.95	-7.42	-27.72
2	13.48	0.98	-5.53
3	8.26	-4.32	-11.33
4	24.93	28.47	113.24

* This is the value of θ^u for this particular transient (see [1]).

The potential energy at this particular θ^u is 0.6261 pu. This energy is calculated with respect to θ^{s2} . The correction term, as per Eq. (4), is

$$V \left| \begin{matrix} \theta^{s2} \\ \theta^{s1} \end{matrix} \right. = -0.0558 \text{ pu}$$

This correction term must be either added to the value of V at θ^u , or subtracted from the value of V at θ^c . In either case, the effect is to reduce the critical transient energy at clearing by about 9% (for this particular case). In other words,

$$V \left| \begin{matrix} \theta^u \\ \theta^{s1} \end{matrix} \right. - V \left| \begin{matrix} \theta^{s2} \\ \theta^{s1} \end{matrix} \right. = V \left| \begin{matrix} \theta^u \\ \theta^{s2} \end{matrix} \right.$$

17-generator system

A three-phase fault at bus 372 (RAUN 345-KV bus) cleared by opening line 372-193 is analyzed. The critical generators are generators No. 5 and 6 (which are close to the fault), as indicated by their u.e.p. values shown in Table II.

TABLE II
17-Generator System (Reduced Iowa System)
Fault at Bus 372, Line 372-193, Cleared

Generator Number	θ^{s1} deg		θ^{s2} deg COI	θ^u deg (for this transient)
	Arbitrary Reference	COI		
1	-27.92	-6.26	-4.92	-1.41
2	-1.37	20.28	22.26	46.63
3	-16.28	5.38	5.60	9.68
4	-26.09	-4.42	-8.34	-23.95
5	-6.24	15.41	18.87	163.55
6	-4.56	17.10	21.57	144.87
7	-23.02	-1.35	-1.53	-15.96
8	-26.95	-5.29	-4.76	-7.98
9	-12.41	9.25	8.91	-6.62
10	-11.12	10.53	12.85	47.77
11	-24.30	-2.64	-1.17	10.28
12	-10.10	11.55	13.90	49.58
13	-28.10	-6.44	-6.61	-25.80
14	-26.76	-5.10	-5.08	-23.62
15	-21.90	0.56	-2.12	-17.61
16	-6.70	14.95	17.78	63.55
17	-4.35	17.30	19.43	50.06

The potential energy at this particular θ^u is 17.16 pu, calculated with respect to θ^{s2} . The correction term is given by

$$V \left| \begin{matrix} \theta^{s2} \\ \theta^{s1} \end{matrix} \right. = -0.498 \text{ pu}$$

Again, this correction term must be added to the potential energy at the u.e.p. (θ^u) if the critical transient energy at clearing is to be determined.

Kinetic Energy Correction

The transient kinetic energy responsible for the separation of the critical generators from the rest of the system is that associated with the gross motion of the critical generators, i.e., that of their inertial center, with respect to the inertial center of the other generators. The remaining portion of the transient kinetic energy need not be absorbed by the network for stability to be maintained. It is identified with the inter-generator motion in each of the groups separating from one another.

The kinetic energy associated with the gross motion of a group of k generators having angular speeds $\omega_1, \omega_2, \dots, \omega_k$ is the same as the kinetic energy of their inertial center. The speed of the inertial center of that group and its kinetic energy are given by

$$\omega = \frac{\sum_{i=1}^k M_i \omega_i}{\sum_{i=1}^k M_i} \quad (5)$$

$$V_{KE} = \frac{1}{2} \left(\sum_{i=1}^k M_i \right) (\omega)^2 \quad (6)$$

If the system contains very large inertias, i.e., the center of inertia of the whole power system is nearly stationary, the kinetic energy given by Eq. (6) tends to separate that group of generators from the rest of the system. If this group is the "critical generators," i.e., the generators tending to separate from the system, then the kinetic energy given by Eq. (6) is the correct V_{KE} contributing to system separation. The kinetic energy term in V_{c1} should be corrected accordingly.

If the system inertias are finite, i.e., if the system's inertial center is not stationary, the kinetic energy that contributes to system separation is a little more complex. Essentially, the disturbance splits the system generators into two groups: the critical generators and the rest of the generators. Their inertial centers have inertia constants and angular speeds M_{cr} , ω_{cr} and M_{sys} , ω_{sys} , respectively. The gross motion of the two groups approximates that of a two-generator system. The kinetic energy causing the separation of the two groups is the same as that of an equivalent one-generator infinite-bus system having inertia constant M_{eq} and angular speed ω_{eq} given by

$$M_{eq} = \frac{M_{cr} M_{sys}}{M_{cr} + M_{sys}} \quad (7)$$

$$\omega_{eq} = (\omega_{cr} - \omega_{sys})$$

and the corresponding kinetic energy is given by

$$V_{KE} = \frac{1}{2} M_{eq} (\omega_{eq})^2 \quad (8)$$

Again the kinetic energy term in V_{c1} should be corrected accordingly.

We note that as the ratio M_{sys}/M_{cr} gets large, Eqs. (6) and (8) give nearly the same result.

The kinetic energy contributing to system separation is illustrated by the following examples.

4-generator system

For a three-phase fault at bus 10 cleared in 0.1s by opening one of the lines 8-10, the critical generator is generator No. 4, which tends to separate from the system. Data for calculating the kinetic energy correction at clearing is shown in Table III.

TABLE III
4-Generator System
Fault at Bus 10, Cleared at 0.1s

Generator Number	M_i	ω^c pu	V_{KE} pu
1	0.1254	-0.0024	0.0507
2	0.034	-0.0005	0.0006
3	0.016	-0.0005	0.0003
4	0.034	0.0095	0.2187

From the data in Table III, we note that the total $V_{KE} = 0.2701$ pu. The kinetic energy of the critical generator (generator No. 4) is 0.2187 pu. The inertia of this machine is about 16% of the total system. While the total inertia of the remaining generators is much greater than that of the critical generator, it is evident that the system is split into two groups: generator No. 4 is accelerating and generators No. 1, 2 and 3 are decelerating. Converting to an equivalent two-generator system, we get

$$M_{cr} = 0.034, \quad \omega_{cr} = 0.00951 \text{ pu}$$

$$M_{sys} = 0.1754, \quad \omega_{sys} = 0.00184 \text{ pu}$$

The one-generator equivalent is given by

$$M_{eq} = 0.02848, \quad \omega_{eq} = 0.01136 \text{ pu}$$

and

$$V_{KE|eq} = 0.2611$$

Comparing this value to the V_{KE} at clearing (0.2703), the kinetic energy correction for this case is -0.009 pu .

17-generator system

For a three-phase fault at bus 372, cleared in 0.15s by opening line 372-193, the critical generators are No. 5 and 6, which tend to separate from the system. Data for calculating the kinetic energy correction at clearing is given in Table IV.

TABLE IV
17-Generator System
Fault at Bus 372, Cleared in 0.15s

Generator Number	M_i	ω^c pu	V_{KE} pu
5	0.08907	0.01142	0.83
6	0.17236	0.02252	6.22
Inertial Center of 5 and 6	0.26143	0.01874	6.53
All others	4.87442	-0.000954	0.92

Although the inertia of the critical generators (No. 5 and 6) is only about 5% of the total inertia of the system, the inertial center of the system is decelerating. To calculate the kinetic energy correction, we compute the data for the equivalent one-generator system as

$$M_{eq} = 0.24812, \quad \omega_{eq} = 0.0197 \text{ pu}$$

and

$$V_{KE|eq} = 6.839$$

From Table II, the total kinetic energy at clearing is 7.97 pu. Thus, the kinetic energy correction at clearing is -1.131 pu .

Finally, we note that the kinetic energy of the inertial center of the critical generators (6.53 pu in Table IV) is somewhat less than the actual kinetic energy contributing to the system separation of 6.839 pu calculated above, even though the critical generators make up only 5% of the system inertia. The reason for this difference is that there is a significant amount of kinetic energy associated with small retarding motion of the large inertia.

Critical Transient Energy at Clearing

The steps for incorporating the above results in predicting critical clearing times are as follows:

1. Find the controlling u.e.p. (θ^u) for the particular transient under consideration.
2. Compute $V_u = V_{\theta^u}^{u2}$
3. Correct for the change in θ^s
corrected $V_u = V_u - V_{\theta^s1}^{s2}$
4. Find conditions at clearing, i.e., θ^c and ω^c .
5. Compute $V_{c1} = V_{\theta^c}^{c1}$
6. Compute kinetic energy correction at clearing ΔV_{KE}
corrected $V_{c1} = V_{c1} - \Delta V_{KE}$
7. For stability, check that
corrected $V_{c1} \leq \text{corrected } V_u$

INVESTIGATION OF CRITICAL CLEARING TIMES

Following the procedure outlined in the previous section, transient stability of the two test systems has been investigated, using the transient energy method, to determine the critical clearing time for each fault considered. The results are then compared to conventional time solutions obtained for the same disturbances.

4-Generator System

The procedure is first illustrated with the 4-generator test system. The disturbance investigated is a three-phase fault at bus 10, cleared by opening line 8-10. For this fault and the resulting post fault network, the corrected potential energy at the controlling u.e.p. has already been calculated

$$V_u (\text{corrected}) = 0.5703 \text{ pu}$$

The critical generator is generator No. 4, and the rest of the system is made up of generators No. 1, 2 and 3.

Several clearing times (t_c) are selected for analysis. For each value of t_c , the corrected value of the fault transient energy V_{c1} is computed. The data is displayed in Table V.

The last column in Table V contains the corrected value of the fault transient energy for different clearing times. This transient energy must not exceed the value of V_{cr} for stability to be maintained. This value was previously given as 0.5704 pu.

The above data indicate that the critical clearing time is between 0.153s and 0.156s, about 0.154s.

Swing curves for some of the cases displayed in Table V are shown in Fig. 3. The rotor position of generator No. 4 and the inertial center for the rest of the generators are given for different clearing times. The time solution data indicate that the system is critically stable for $t_c = 0.156s$,

TABLE V
Fault Transient Energy
4-Generator System, Fault at Bus 10

t_c s	ω_4 pu	ω_{sys} pu	ω_{eq}	$V_{KE eq}^*$	$V_{KE c1}$ pu	ΔV_{KE}	V_{c1}	Corrected V_{c1} pu
0.1	0.00951	-0.00184	0.01136	0.2611	0.2701	-0.0090	0.2166	0.2076
0.148	0.01406	-0.00273	0.016786	0.5703	0.5859	-0.0156	0.5414	0.5258
0.150	0.01430	-0.00278	0.01708	0.59043	0.6057	-0.0153	0.5602	0.5450
0.153	0.0145	-0.002786	0.017286	0.6048	0.6256	-0.0208	0.5879	0.5671
0.156	0.0148	-0.00286	0.01767	0.6300	0.6499	-0.0199	0.6166	0.5967
0.159	0.0151	-0.00295	0.01805	0.6593	0.6748	-0.0155	0.6461	0.6306

* $M_{eq} = 0.02848 \text{ pu}$.

which is only 0.02s higher than the value of the critical clearing time predicted by the direct method.

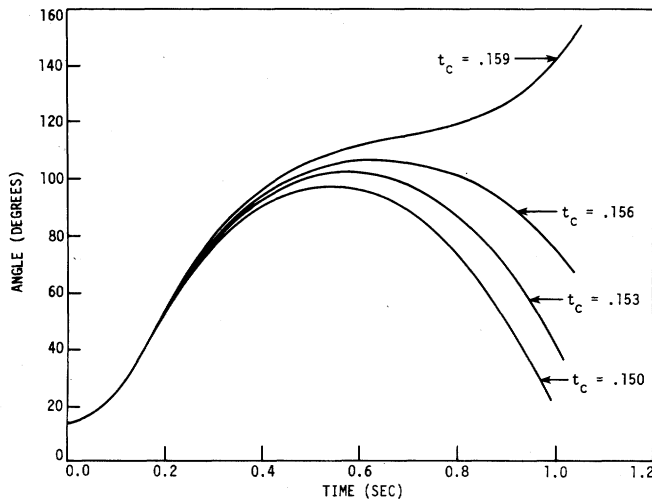


Fig. 3. 4-generator test system. Swing curves of generator No. 4 for a fault at Bus 10; $t_c = 0.150, 0.153, 0.156$ and 0.159 s.

17-Generator System

For this system, the following results are investigated:

- A three-phase fault at RAUN (bus 372), cleared by opening line 372-193.
- A three-phase fault at Council Bluffs (C.B.), unit No. 3 (bus 436), cleared by opening line 436-771.
- A three-phase fault at Ft. Calhoun (bus 773), cleared by opening line 773-779.
- A three-phase fault at Cooper (bus 6), cleared by opening line 6-439.

Critical Potential Energy

Using a special computer program, the controlling u.e.p. for each disturbance is determined. This particular u.e.p., which assumes a certain mode of instability, identifies the so-called critical generators, i.e., the generators tending to separate from the rest of the system by the disturbance. This information is checked against time solutions to correctly identify the critical generators. In some cases, several modes of instability were found to be possible for the same initial disturbance, which made the identification of the correct controlling u.e.p. difficult.

Once the critical generators and the controlling u.e.p. are identified the corresponding value of transient energy function is computed. This value is then corrected to give the critical potential energy for this disturbance (against which the fault transient energy is to be compared). This information is displayed in Table VI.

TABLE VI
Corrected Critical Potential Energy
17-Generator System

Fault	Critical Generators	$V_u \left \frac{\partial u}{\partial s} \right ^2$	Corrected V_u pu
Raun	5,6	17.162	16.664
Council Bluffs No. 3	10,12	12.025	12.301
Ft. Calhoun	16,2,10,12,17	28.016	28.250
Cooper	2	11.365	11.558

The data used in Table VI for the Ft. Calhoun fault represent one of several modes of instability for this disturbance. Each mode represents a

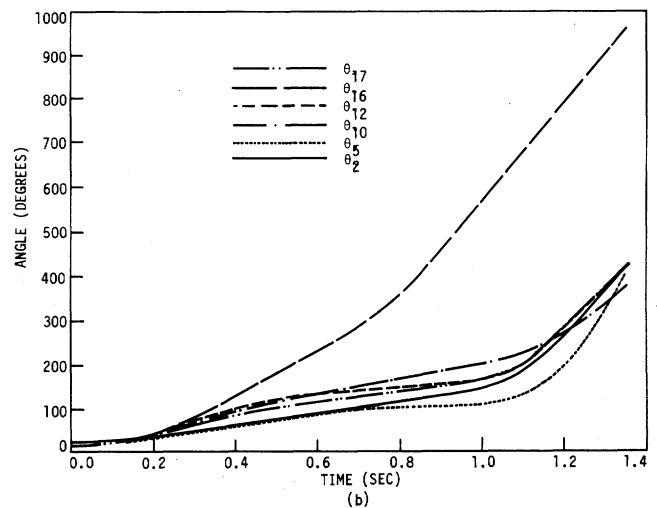
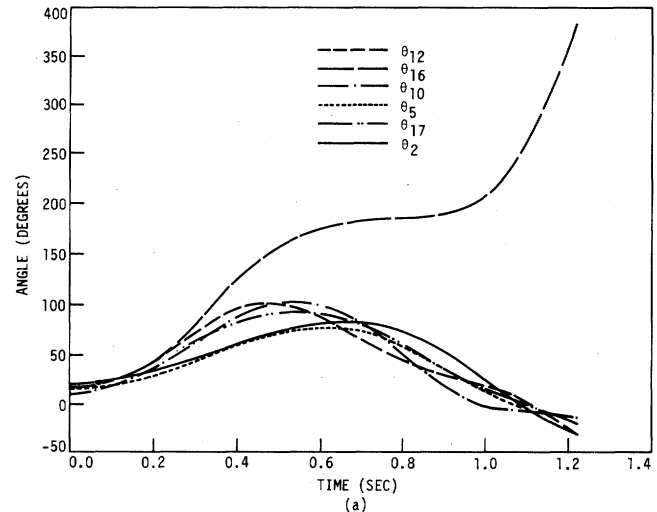


Fig. 4. 17-generator test system. Swing curves of generators No. 2, 5, 10, 12, 16 and 17 for fault at Ft. Calhoun; (a) $t_c = 0.357$ s, (b) $t_c = 0.423$ s.

certain group of generators, including generator No. 16, which is close to the fault, separating from the rest of the system. For each of these modes, a controlling u.e.p. is identified, with a corresponding critical potential energy. Thus, there is a cluster of u.e.p.'s with energy levels of similar magnitudes, i.e., in the range of 25.5-28.5 pu. The difficulty in identifying the particular u.e.p. of interest is illustrated in Fig. 4, which shows the swing curves for generators No. 2, 5, 10, 12, 16 and 17 for different clearing times, each sufficiently large to drive the system unstable.

The data in Fig. 4 indicate that the disturbance tends to separate generators No. 2, 5, 6, 10, 12, 16 and 17 from the rest of the system. The energy level needed to separate this group is quite high (about 30 pu). As the system trajectory moves toward the u.e.p. for this group, it encounters the cluster of u.e.p.'s with potential energy levels of 25.5-28.5 pu. This cluster controls the first swing stability of the system for this disturbance. This is confirmed by the fact that the system trajectory in the critically unstable case ($t_c = 0.357$ s) acquires a maximum potential energy of about 25.9 pu.

This particular mode of instability for the Ft. Calhoun fault is chosen as the most likely one because the rotor positions and the energy level are closest to the results obtained by time solutions.

Fault Transient Energy

The corrected values of the fault transient energy for different values

TABLE VII
Fault Transient Energy
17-Generator System

t_c s	ω_{cr} pu	ω_{sys} pu	ω_{eq} pu	$V_{KE eq}$ pu	$V_{KE cl}$ pu	ΔV_{KE} pu	V_{cl} pu	Corrected V_{cl} pu
a) Raun fault: critical generators 5,6;								
	$M_{5,6} = 0.2614$ pu, $M_{sys} = 4.8744$ pu, $M_{eq} = 0.24812$							
0.15	0.01875	-0.00101	0.01976	6.8796	7.999	-1.119	10.286	9.167
0.189	0.02336	-0.00112	0.02448	10.5678	12.369	-1.801	18.0703	16.269
0.192	0.02371	-0.00127	0.02498	11.003	12.742	-1.740	18.769	17.029
0.1924	0.02373	-0.00127	0.02500	11.019	12.793	-1.774	18.890	17.116
0.1932	0.02387	-0.00127	0.025414	11.147	12.892	-1.745	19.054	17.309
b) Council Bluffs No. 3 fault: critical generators 10,12;								
	$M_{cr} = 0.1315$ pu, $M_{sys} = 5.00385$ pu, $M_{eq} = 0.12813$ pu							
0.15	0.02001	-0.000505	0.020505	3.829	6.326	-2.497	8.450	5.953
0.20	0.026485	-0.000696	0.027181	6.727	10.468	-3.741	16.411	12.670
0.204	0.02704	-0.00072	0.027764	7.019	10.826	-3.807	17.167	13.360
0.206	0.027224	-0.00072	0.027944	7.111	11.007	-3.896	17.550	13.653
c) Ft. Calhoun fault: critical generators 2,10,12,16,17;								
	$M_{cr} = 0.57614$ pu, $M_{sys} = 4.55971$ pu, $M_{eq} = 0.51151$ pu							
0.30	0.01212	-0.00153	0.01365	6.774	9.9185	-3.145	20.902	17.757
0.350	0.012992	-0.001642	0.014634	7.784	11.722	-3.938	27.812	23.874
0.357	0.013229	-0.001672	0.014901	8.071	11.980	-3.909	28.802	24.893
0.400	0.014066	-0.001777	0.015844	9.125	13.651	-4.526	34.870	30.344
d) Cooper fault: critical generator 2;								
	$M_{cr} = 0.18334$ pu, $M_{sys} = 4.95251$ pu, $M_{eq} = 0.1768$							
0.150	0.01530	-0.00057	0.01587	3.163	6.257	-3.094	8.093	4.999
0.204	0.02080	-0.00077	0.02157	6.062	10.827	-4.765	16.528	11.763
0.216	0.02200	-0.00081	0.02281	6.540	11.951	-5.411	18.856	13.444
0.220	0.02240	-0.00083	0.02323	6.780	12.333	-5.553	19.665	14.112

of clearing time (t_c) are computed. The results for the faults at RAUN, C.B. No. 3, Ft. Calhoun, and Cooper are displayed in the last column of Table VII.

Critical Clearing Time (t_c)

The critical clearing time for each fault is determined from the data in Tables VI and VII.

For the RAUN fault, the results in Table VII show that the critical clearing time, which gives a fault transient energy equal to $V_{cr} = 16.664$ pu, is about 0.191s. Swing curves for different clearing times of this disturbance are shown in Fig. 5, from which we know that the swing curve for $t_c = 0.1923$ is critically stable. Therefore, the critical clearing time predicted by the direct method compares favorably with that obtained by time solutions.

For the fault at C.B. No. 3, the predicted critical clearing time, having a fault transient energy equal to $V_{cr} = 12.301$ pu, is just under 0.20s. Time solutions, shown in Fig. 6, give a critical clearing time of just under 0.204s.

For the Ft. Calhoun fault, the predicted critical clearing time, having a fault transient energy equal to $V_{cr} = 28.25$ pu, is about 0.382s. Time solutions, shown in Fig. 7, give a critical clearing time just under 0.357s.

For the fault at Cooper, the predicted critical clearing time, having a fault transient energy equal to $V_{cr} = 11.558$ pu, is about 0.204s. Time solutions, shown in Fig. 8, give a critical clearing time of about 0.216s.

DISCUSSION AND CONCLUSIONS

The results presented in this paper clearly show that first swing transient stability of a multi-generator power system can be accurately predicted by direct methods of transient stability analysis. The success of

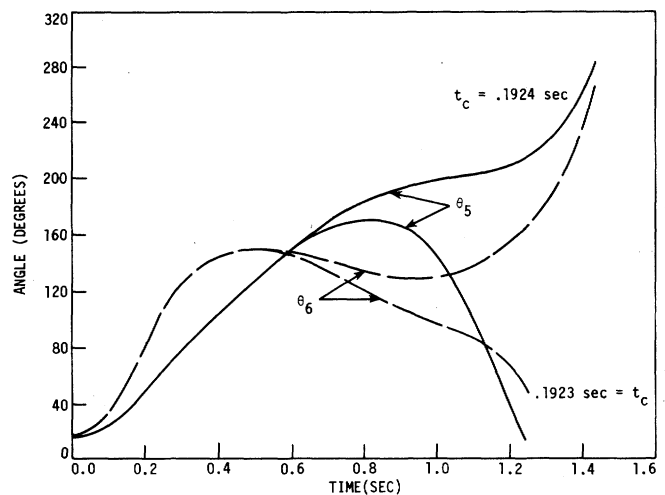


Fig. 5. 17-generator test system. Swing curves of generators No. 5 and 6 for RAUN fault; $t_c = 0.1923$ and 0.1924 s.

these methods depends upon:

- Use of a function that accurately describes the system transient energy.
- Identification of the mode of instability of the system for the disturbance under consideration and the controlling u.e.p. for this disturbance.
- Accurate computation of the fault transient energy contributing to system separation.

The above statements are substantiated by the data shown in Tables VI

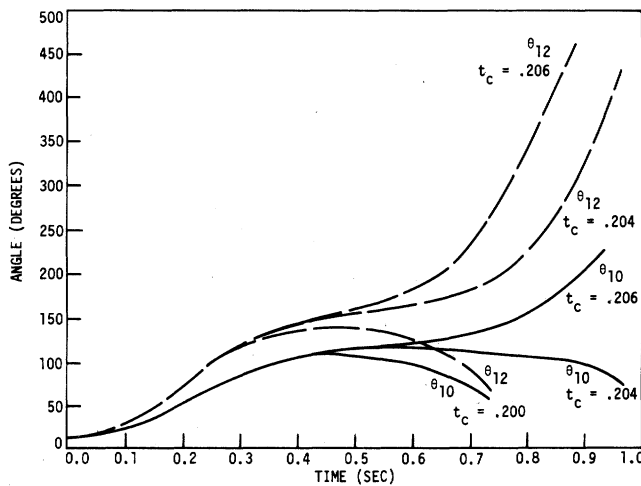


Fig. 6. 17-generator test system. Swing curves of No. 10 and 12 for fault at Council Bluffs No. 3; $t_c = 0.200, 0.204, \text{ and } 0.206\text{s}$.

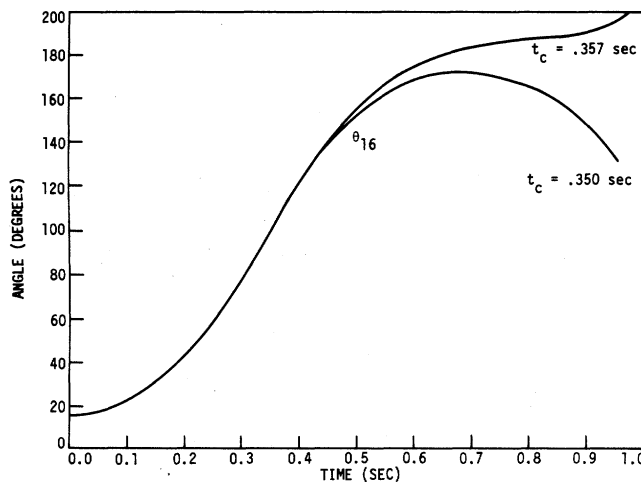


Fig. 7. 17-generator test system. Swing curves of generator No. 16 for Ft. Calhoun fault; $t_c = 0.350, 0.357\text{s}$.

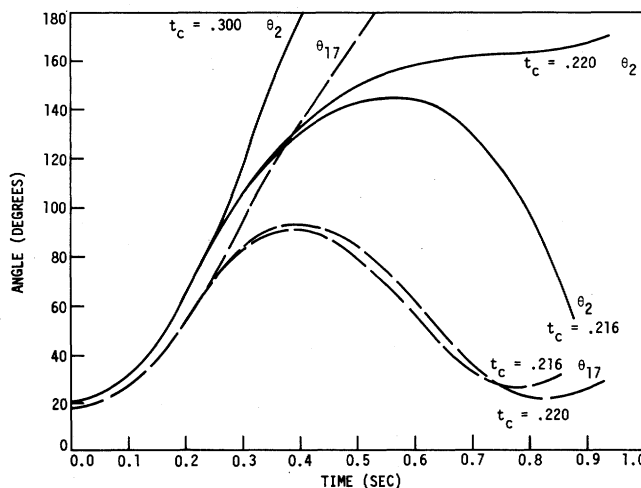


Fig. 8. 17-generator test system. Swing curves of generator No. 2 for fault at Cooper; $t_c = 0.215, 0.220, 0.30\text{s}$.

and VII. For example, the predicted values of critical clearing times for the four cases investigated on the 17-generator system compare quite favorably with the data obtained by time solutions, even for the most complex mode of instability in the case of the Ft. Calhoun fault (7% error).

The importance of the correction for the fault transient energy that does *not* contribute to system separation is clearly evident from the data shown in Table VII. If the procedure commonly suggested in the literature is adopted, the values of V_{c1} (shown in column 7) would be used. These values would predict critical clearing times that are grossly in error (as much as 30% error in critical energy and 24% error in t_{cr}).

The procedure developed in the research reported in this paper has been successfully applied to a realistic, and rather complex, test power network. It has, therefore, clearly demonstrated that direct methods can be reliably used for first swing transient stability analysis. These methods will add a powerful tool which is potentially valuable to both the system planner and the system operator. The system planner would have information on the robustness of the power system when subjected to disturbances, and, in planning studies, would be able to substantially reduce the number of stability cases to analyze in detail. To the system operator, these methods offer the prospect of having on-line transient stability analysis, and the potential for having a tool for dynamic security assessment.

ACKNOWLEDGEMENT

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The authors wish to express their thanks to Glenn Walkup and Lloyd Sievers of Iowa Power and Light Company for their helpful comments and suggestions.

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3. A. A. Fouad, "Stability Theory — Criteria for Transient Stability." Proc. Eng. Foundation Conference on Systems Engineering for Power. Publication No. CONF-750867, pp. 421-50, Henniker, NH, 1975.

Combined Discussion^{1,2}

M. Ribbens-Pavella and P. G. Murthy (University of Liege, Liege, Belgium): The authors present an interesting contribution within the framework of the practical application of the transient energy function to transient stability analysis of power systems. Indeed, whether based on energy functions or on Liapunov functions, the success of direct methods depends strongly upon the suitable identification of the "controlling or appropriate" u.e.p. that can provide fairly accurate practical results with minimum computational effort. A few remarks could be useful; for the sake of brevity we will focus upon only the two following points, brought up in the "Discussion and Conclusions" Section of the Part II paper.

We agree with the authors' statement according to which "the procedure developed in the research reported in this paper has been successfully applied to a realistic, and rather complex, test power network". We believe, however, that from there on arriving at the conclusion according to which "it has, therefore, clearly demonstrated that direct methods can be reliably used for first swing transient stability analysis" may be a slight overstatement; indeed this conclusion is based only on a few disturbance simulations on the 17-machine and the 4-machine systems.

The second point is related to the last sentence of the "Discussion and Conclusions" Section: "... these methods offer the prospect of having on-line transient stability analysis...". This may perhaps be a somewhat misleading statement. Indeed, we believe that by its very nature the conventional procedure is essentially an off-line one, since it consists in characterizing the behaviour of a system from its transient stability point of view in terms of the magnitude of the critical clearing time related to a specified disturbance; moreover, this has to be repeated for many disturbances at different operating points. This is true, no matter whether direct method or standard step-by-step integration technique is used. It could therefore be useful if the authors would clarify their thought when speaking about "on-line transient stability analysis".

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A. A. Fouad and S. E. Stanton: The authors wish to thank professors Ribbens-Pavella and Murthy for their discussion. Concerning the two issues they raised, we wish to make the following comments:

1. Our conclusion that "direct methods can be reliably used for first swing transient stability analysis" is based on *numerous* (perhaps hundreds) disturbance simulations on the 17-machine and the 4-machine systems [1]. Because of the space limitations, however, only a few simulations are presented in the paper.

The 17-machine system represents a reduced version of a much larger network of the State of Iowa. Its first swing transient response was validated (by time simulations) against the response of the larger network, even with the excitation system represented on most of the generators in the large network. This system with seven generators located electrically close to each other (at the Western part of the state) has been studied for disturbances near these generators. The transient behavior, and the possible modes of system separation, are exceedingly complex. In our judgement, the fact that these complexities could be properly accounted for by the methods proposed in the paper, supports our conclusion.

2. Our statement that "...these methods offer the prospect of having on-line transient stability analysis..." is far from being misleading. It means that these methods have the *potential* of performing first swing transient analysis at or near real time, i.e., in seconds. The objective would be to perform fast dynamic security assessment for the benefit of the system operator. The operator would then be "alerted" to the system's vulnerability to certain contingencies, in order that corrective actions may be taken if deemed necessary. We did *not* have in mind determination of critical clearing times. This is discussed in the reference below.

REFERENCE

- [1] A. A. Fouad et al, "Transient Stability Margin as a Tool for Dynamic Security Assessment". EPRI Report EL-1755 (under publication in April 1981).

Manuscript received March 31, 1981.

¹ A. A. Fouad and S. E. Stanton, Transient Stability of a Multi-Machine Power System, Pt. I: Investigation of System Trajectories, this issue, pp.3408-3416.

² A. A. Fouad and S. E. Stanton, Transient Stability of a Multi-Machine Power Systems, Pt. II: Critical Transient Energy, this issue, pp. 3417-3424.